

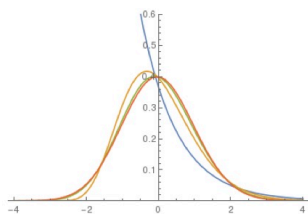
EXERCISE 3.1: CENTRAL LIMIT THEOREM (CLT)

(6P)

Let $X_1, X_2, \dots, X_N$ be statistically independent and identically distributed random variables with the probability density function

$$p_X(x) = \begin{cases} e^{-x} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

With this exercise we would like to demonstrate the CLT for $Z_N = \sum_{i=1}^N X_i$ in the limit $N \rightarrow \infty$.



- (a) Show that the N -fold convolution product of $p(x)$ is given by (1P)

$$p_{Z_N}(z) = p_X^{*N}(z) = \underbrace{(p_X * p_X * \dots * p_X)}_{N \text{ times}}(z) = \frac{z^{N-1}}{(N-1)!} e^{-z} \quad (z \geq 0).$$

- (b) Check the norm and compute the mean and the variance of $p_{Z_N}(z)$. (1P)

- (c) Standardize the distribution by shifting and rescaling Z_N in such a way that the new random variable $\tilde{Z}_N$ has a probability density $p_{\tilde{Z}_N}(z)$ with unit norm, zero mean, and unit variance (see figure). (1P)

- (d) Show that the cumulant-generating function $K_{\tilde{Z}_N}(t)$ of the standardized probability density $p_{\tilde{Z}_N}(z)$ is given by (1P)

$$K_{\tilde{Z}_N}(t) = \frac{1}{2} N \ln N - t\sqrt{N} - N \ln(\sqrt{N} - t).$$

- (e) Compute all cumulants κ_n and show that only κ_2 survives in the limit $N \rightarrow \infty$. What does it mean? (2P)

c)

$$\tilde{Z} = \frac{Z_N - N}{\sqrt{N}} = \frac{Z_N - E}{\sqrt{\text{Var}}}$$

* shift from N to origin

* reformation of the width with standard deviation $\sqrt{N}$

$$d) \mu_{\tilde{Z}_N} = E(e^{t\tilde{Z}_N}) =$$

$$= E\left(e^{t \frac{Z_N}{\sqrt{N}}} \cdot e^{-t \frac{N}{\sqrt{N}}}\right) =$$

$$= e^{-t\sqrt{N}} E\left(e^{t\sqrt{N} \frac{Z_N}{N}}\right)$$

$$= e^{-t\sqrt{N}} \mu_{Z_N}\left(\frac{t}{\sqrt{N}}\right)$$

$$K_{\tilde{Z}_N} = \ln(\mu_{\tilde{Z}_N}) = -t\sqrt{N} + \ln(\mu_{Z_N}\left(\frac{t}{\sqrt{N}}\right))$$

$$\begin{aligned} \mu_X(t) &= E(e^{tx}) = \int_0^\infty e^{tx} e^{-x} dx = \\ &= \int_0^\infty e^{(t-1)x} dx = \left[\frac{e^{(t-1)x}}{-(t-1)} \right]_0^\infty = \\ &= \frac{1}{1-t} \rightarrow \mu_{Z_N} = \left(\frac{1}{1-t}\right)^N = (1-t)^{-N} \end{aligned}$$

$$\Rightarrow \mu_{Z_N}\left(\frac{t}{\sqrt{N}}\right) = \left(1 - \frac{t}{\sqrt{N}}\right)^{-N}$$

$$K_{\tilde{Z}_N} = -t\sqrt{N} - N \ln\left(1 - \frac{t}{\sqrt{N}}\right)$$

$$= -t\sqrt{N} - N \left(\ln(\sqrt{N}-t) - \ln\sqrt{N} \right)$$

$$= \frac{1}{2} N \ln(N) - t\sqrt{N} - N \ln(\sqrt{N}-t)$$

$$e) x_n = \frac{d^n K_{\tilde{Z}_N}(t)}{dt^n} \Big|_{t=0}$$

$$x_1 = -\sqrt{N} - \frac{N}{\sqrt{N}-t} \cdot (-1) \Big|_{t=0} = 0$$

$$x_2 = \frac{d}{dt} \frac{N}{\sqrt{N}-t} \Big|_{t=0} = \frac{-N \cdot (-1)}{(\sqrt{N}-t)^2} \Big|_{t=0} = \frac{N}{N} = 1$$

$$\begin{aligned} x_3 &= \frac{d}{dt} \frac{N}{(\sqrt{N}-t)^2} \Big|_{t=0} = \frac{-N \cdot 2(\sqrt{N}-t)(-1)}{(\sqrt{N}-t)^4} \Big|_{t=0} = \\ &= \frac{2N^{3/2}}{N^2} = 2 \frac{1}{\sqrt{N}} \rightarrow 0 \text{ for } N \rightarrow \infty \end{aligned}$$

Equally for $n=4, 5, \dots$

For $N \rightarrow \infty$, we obtain $x_2=1$ and $x_n=0$ for $n \neq 2$, which is characteristic for a gaussian distribution.

a) Compute some convolution products:

$$\begin{aligned} p_X^{*2}(z) &= \int_{-\infty}^{\infty} e^{-x} e^{-(z-x)} dx = \\ &= \int_0^z e^{-z} dx = z \cdot e^{-z} = \frac{z^{2-1}}{(2-1)!} e^{-z} \end{aligned}$$

$$\begin{aligned} p_X^{*3}(z) &= \int_0^z x \cdot e^{-x} e^{-(z-x)} dx \\ &= e^{-z} \int_0^z x dx = \frac{1}{2} z^2 e^{-z} = \frac{z^{3-1}}{(3-1)!} e^{-z} \end{aligned}$$

$$\begin{aligned} p_X^{*(N+1)}(z) &= \int_0^z \frac{x^{N-1}}{(N-1)!} e^{-x} e^{-(z-x)} dx = \\ &= e^{-z} \int_0^z \frac{x^{N-1}}{(N-1)!} dx = \frac{z^N}{N!} e^{-z} \\ &= \frac{z^{(N+1)-1}}{((N+1)-1)!} e^{-z} \end{aligned}$$

The formula seems to be valid for some examples and for the generalization for N

$$\begin{aligned} b) \int_0^\infty \frac{z^{N-1}}{(N-1)!} e^{-z} dz &= \frac{1}{(N-1)!} \int_0^\infty z^{N-1} e^{-z} dz \\ &= \frac{(N-1)!}{(N-1)!} = 1 \rightarrow \text{normalization fulfilled} \end{aligned}$$

$$\begin{aligned} E &= \int_0^\infty z \cdot \frac{z^{N-1}}{(N-1)!} e^{-z} dz = \frac{1}{(N-1)!} \int_0^\infty z^N e^{-z} dz \\ &= \frac{\Gamma(N+1)}{(N-1)!} = \frac{N!}{(N-1)!} = N \end{aligned}$$

$$m_2 = \int_0^\infty z^2 \frac{z^{N-1}}{(N-1)!} e^{-z} dz = \frac{1}{(N-1)!} \Gamma(N+2) = \frac{(N+1) \cdot N}{(N-1)!}$$

$$\text{Var} = E(Z_N^2) - E(Z_N)^2 = N$$

Physics of Complex Systems: Exercise 3

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May 5, 2026

Task 3.1: Central Limit Theorem

(a)

(b)

(c) *see page one*

(d)

(e)

Task 3.2: Vector notation and Liouville operator

(a)

$$\dot{P}_c(t) = \sum_{c'} P_{c'}(t) \omega_{c' \rightarrow c} - P_c(t) \sum_{c''} \omega_{c \rightarrow c''} \quad (1)$$

to write the whole term as a sum over c' , we use the Kronecker delta $\delta_{c',c}$. With $\delta_{c',c} P_c = \delta_{c',c} P_{c'}$ we get:

$$\dot{P}_c(t) = \sum_{c'} P_{c'}(t) \omega_{c' \rightarrow c} - \sum_{c'} \delta_{c',c} \left(\sum_{c''} \omega_{c \rightarrow c''} \right) P_{c'}(t) \quad (2)$$

$$= \sum_{c'} \left(\omega_{c' \rightarrow c} - \delta_{c',c} \sum_{c''} \omega_{c \rightarrow c''} \right) P_{c'}(t) \quad (3)$$

Now we expand at the Operator representation of the probability vector:

$$|P\rangle = \sum_{c'} P_{c'}(t) |c'\rangle \quad (4)$$

$$\Rightarrow \langle c | \dot{P} \rangle = - \langle c | \mathcal{L} | P \rangle = - \sum_{c'} \langle c | \mathcal{L} | c' \rangle P_{c'}(t) \quad (5)$$

$$= \dot{P}_c(t) \quad (6)$$

A comparison of both results shows:

$$- \langle c | \mathcal{L} | c' \rangle = \omega_{c' \rightarrow c} - \delta_{c',c} \sum_{c''} \omega_{c \rightarrow c''} \quad (7)$$

no we relabel the indices ($c \rightarrow c'$, $c' \rightarrow c$) and get the expression we should prove:

$$\langle c | \mathcal{L} | c' \rangle = -\omega_{c \rightarrow c'} - \delta_{c,c'} \sum_{c''} \omega_{c \rightarrow c''} \quad (8)$$

(b)

For the given transition rates, the Liouville Operator is:

$$\mathcal{L} = \begin{pmatrix} 3.5 & -1 & 0 \\ -3 & 3 & -3 \\ -0.5 & -2 & 3 \end{pmatrix} \quad (9)$$

(c)

With the following code we calculated for the right eigenvector $(0.4, 1.4, 1)^T$ and for the left eigenvector $(1, 1, 1)$

```
import sympy as sp
# Define the matrix
M = sp.Matrix([ [3.5, -1, 0],
                 [-3, 3, -3],
                 [-0.5, -2, 3]])
# Calculate eigenvectors for eigenvalue 0
right_eigenvectors = M.nullspace()
left_eigenvectors = M.T.nullspace()

sp.pprint(right_eigenvectors)
sp.pprint(left_eigenvectors)
```

(d)

$$e_r = \begin{pmatrix} 0.4 \\ 1.4 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{3.12}} \approx \begin{pmatrix} 0.226 \\ 0.793 \\ 0.566 \end{pmatrix} \quad (10)$$